

Propylene Glycol - Cumberland River Compact

**Category: Propylene Glycol Aquatic Chemistry & River Spill Dangers |
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Source Content

Propylene glycol is a small, carbon-based molecule that is in the alcohol family, along with the more familiar compounds ethanol (drinking alcohol) and isopropyl (rubbing alcohol). Propylene glycol is a common ingredient in pharmaceuticals, cosmetics and deodorant sticks. It also finds application as a de-icing compound and antifreeze. Propylene Glycol is an organic contaminate in the same family chemicals like PCBs, DDT, chlordane, and dioxins, which are listed by the EPA as probable human carcinogens (cancer causing agents) and that tend to remain in the environment for an extremely long time. In some waterbodies, these substances have accumulated in sediment and pose a health threat to those that consume fish or shellfish. Propylene glycol is known to exert high levels of biochemical oxygen demand (BOD) during degradation in surface waters. This process can adversely affect aquatic life by consuming oxygen needed by aquatic organisms for survival. Problematically unhealthy levels of organic contaminants can be below current detection levels. Detection of these substances is generally made either by analyzing fish tissue levels and/or by use of sediment screening values provided by the EPA. Since organic contaminants can bioaccumulate in fish, it is important to make sure catfish and other species consumed by people are safe to eat. Children and pregnant or nursing people are the most sensitive sub- population. Typically, propylene glycol finds its way to our water via production and processing facility wastewater, from spills, or from de-icing fluid carried by stormwater runoff.